PROBLEM 14

Conservative Force and Potential V

Original statement · Problems.pdf, p. 9

Phase Analysis

A particle of mass m is moving along the Ox axis under the force $\mathbf{F}(x) = mKxe^{x/a}\mathbf{i}$, where K and a are known positive constants. The initial position of the particle is $x = a$ and the initial velocity is $\mathbf{v} = -v_0\mathbf{i}$. Describe the possible types of motion of the particle as a function of v_0.

Worked Solution (4-Phase Methodology)

PHASE 1

Constraints, Degrees of Freedom & Coordinates

- **System:** a particle of mass m on the Ox axis under $\mathbf{F}(x) = mKxe^{x/a}\,\mathbf{i}$ ($K > 0, a > 0$), starting at $x(0) = a$ with $\dot{x}(0) = -v_0$ ($v_0 > 0$).
- **Constraints:** motion along the axis (smooth guide); $\mathrm{CDOF} = 1$, coordinate x.
- **Parameters:** m [kg], K [s^{-2}], a [m], v_0 [m/s].

PHASE 2

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0 = \{O; \mathbf{i}, \mathbf{j}, \mathbf{k}\}$ with $\mathbf{i}$ along the axis.

PHASE 3

Derivation

3.1 Kinematics

$$\mathbf{r} = x\,\mathbf{i}, \qquad \mathbf{v} = \dot{x}\,\mathbf{i}, \qquad \mathbf{a} = \ddot{x}\,\mathbf{i}$$

3.2 Forces

- **Applied force:** $\mathbf{F}(x) = mKxe^{x/a}\,\mathbf{i}$, which depends only on x and is therefore conservative. **Constraint forces:** perpendicular to Ox. **Inertia forces:** none.
- **Potential by integration by parts.** $V(x) = -\int F\,dx = -mK\int xe^{x/a}\,dx$. With $u = x$, $du = dx$, $dv = e^{x/a}dx$, $v = ae^{x/a}$, and $\int u\,dv = uv - \int v\,du$:

$$\int xe^{x/a}dx = axe^{x/a} - \int ae^{x/a}dx = axe^{x/a} - a^2e^{x/a} = a^2e^{x/a}\left(\frac{x}{a} - 1\right)$$

$$V(x) = -mKa^2e^{x/a}\left(\frac{x}{a} - 1\right) \qquad (C = 0 \text{ in the primitive, so that } V(a) = 0)$$

- **Landscape.** $\dfrac{dV}{dx} = -F = -mKxe^{x/a}$ vanishes only at $x = 0$. The second derivative, with the product rule, is $\dfrac{d^2V}{dx^2} = -mKe^{x/a}\left(1 + \dfrac{x}{a}\right)$, so $V''(0) = -mK < 0$: a local maximum with $V(0) = mKa^2$ (barrier). Also $V(-\infty) = 0$ (since $e^{x/a} \to 0$) and $V \to -\infty$ as $x \to +\infty$.

3.3 Equations of Motion

- **Newton's second law:** $m\ddot{x} = mKxe^{x/a}$. The energy is a first integral: $E = \frac{1}{2}m\dot{x}^2 + V(x)$. At $t = 0$, $V(a) = 0$, so

$$E = \tfrac{1}{2}mv_0^2$$

3.4 Integration and Answers

Critical velocity

> **Pedagogical Justification:** the particle starts at $x = a > 0$ moving toward the barrier at $x = 0$. It can reach the top of the barrier only if its energy is at least the barrier height, $E \geq V(0)$:

$$\tfrac{1}{2}mv_{0,\text{crit}}^2 = mKa^2 \implies v_{0,\text{crit}} = a\sqrt{2K}$$

Classification of the motion

- **Regime 1,** $v_0 < a\sqrt{2K}$ **(**$E < V(0)$**).** The kinetic energy is not enough to overcome the barrier: V is strictly decreasing on $(0, a)$ ($V' = -mKxe^{x/a} < 0$) from $V(0) = mKa^2 > E$ down to $V(a) = 0 < E$ (since $v_0 > 0$), so there is exactly one point $x_{\text{turn}} \in (0, a)$ with $V(x_{\text{turn}}) = E$. For $x > 0$ the force is positive, $F = mKxe^{x/a} > 0$, so the particle decelerates while approaching the origin, stops at a turning point $x_{\text{turn}} \in (0, a)$ where $V(x_{\text{turn}}) = E$, and is accelerated back toward $+x$ ($F(x_{\text{turn}}) > 0$). It passes $x = a$ and moves toward $+\infty$.
- **Regime 2,** $v_0 = a\sqrt{2K}$ **(**$E = V(0)$**).** The energy equals the barrier height. Near $x = 0$, $V(x) \approx V(0) + V'(0)x + \tfrac{1}{2}V''(0)x^2 = mKa^2 - \tfrac{1}{2}mKx^2$, so with $E = V(0)$:

$$\dot{x} = -\sqrt{\frac{2}{m}\left[V(0) - V(x)\right]} \approx -\sqrt{\frac{2}{m} \cdot \frac{1}{2}mKx^2} = -\sqrt{K}\,x$$

Separating variables, the time to go from a point x_1 already close to the origin (where the quadratic approximation holds) to a point x nearer still follows from Barrow's rule, measuring t from the instant at x_1: $\int_{x_1}^{x} \frac{dx'}{x'} = -\sqrt{K}\int_0^t dt'$, i.e. $\left[\ln x'\right]_{x_1}^{x} = \ln\frac{x}{x_1} = -\sqrt{K}\,t$, i.e. $t = \frac{1}{\sqrt{K}}\ln\frac{x_1}{x}$, which diverges as $x \to 0$. The particle needs **infinite time** to reach $x = 0$ (it approaches it asymptotically; the barrier top is an unstable equilibrium).

- **Regime 3,** $v_0 > a\sqrt{2K}$ **(**$E > V(0)$**).** The particle reaches $x = 0$ with the speed $\sqrt{\frac{2}{m}\left[E - V(0)\right]} = \sqrt{v_0^2 - 2Ka^2} > 0$. For $x < 0$ the force $F = mKxe^{x/a}$ is negative, so it keeps accelerating toward $-x$ and goes to $-\infty$ (as $x \to -\infty$, $V \to 0$ and the speed tends to v_0, the speed it had at $x = a$ because $V(a) = 0$ too).

PHASE 4

Interpretation & Checks

- **Dimensional check:** $[v_{0,\mathrm{crit}}] = [a]\sqrt{[K]} = \mathrm{m}\cdot\mathrm{s}^{-1}$; $[mKa^2] = \mathrm{kg\,s^{-2}m^2} = \mathrm{J}$, like $\tfrac{1}{2}mv_0^2$.
- **Limits:** for $K \to 0$ the force vanishes, $v_{0,\mathrm{crit}} \to 0$ and every $v_0 > 0$ crosses the origin (free particle); for $v_0 \gg a\sqrt{2K}$ the barrier is negligible and $v(0) \to v_0$.
- **Check against the official key:** the key of `Problems.pdf` states: (written there as $a\sqrt{2k}$, with k the constant K of the statement) if $v_0 > a\sqrt{2K}$ the particle moves toward $-\infty$; if $v_0 < a\sqrt{2K}$ it moves toward $+\infty$ after decelerating before $x = 0$; if $v_0 = a\sqrt{2K}$ it takes infinite time to reach $x = 0$. These are the three regimes above.
- **Computational check (extension, not a course tool)** (DOP853, $m = K = a = 1$, $v_{0,\mathrm{crit}} = 1.41421$): for $v_0 = 1.2$ the particle turns at $x = 0.60801$ (the root of $-e^x(x-1) = 0.72$ is 0.608007) and returns to $+\infty$; for $v_0 = 1.6$ it crosses the origin; for $v_0 = 0.999999\,v_{0,\mathrm{crit}}$ it turns at $x = 0.002$ very close to the top of the barrier.

Source: Problems.pdf